

Alphanumeric LED display unit

Berlin Television Electronics Factory

Specifications ($\delta a = 25^\circ\text{C}$)

		min.	typ.	max.	
Luminous intensity ¹ 2 at $U_{1z} = 5\text{ V}$	I	25	—	—	μcd
Luminous intensity ratio ¹ 2 3 from diode to diode point at $U_{1z} = 5\text{ V}$	I	—	—	2.8	
Wavelength of maximum spectral emission ⁴	V_{max} V_{min}	630	665	690	nm
H input current at $U_{\text{cc}} = 5.25\text{ V}$, $U_{1H} = 2.4\text{ V}$	I_{IH}	—	—	0.08	mA
L-input current data	I_{LC}	—	—	2	mA
L input current cycle at $U_{\text{cc}} = 5.25\text{ V}$, $U_{1L} = 0.4\text{ V}$	I_{LD} -1	—	—	0.8	mA
Input diode voltage at $U_{\text{cc}} = 4.75\text{ V}$, $-I_c = 12\text{ mA}$	$-U_{IC}$	—	—	1.5	V
Line input current at $U_{1z} = 5\text{ V}$	I_{1z}	—	—	500	mA
Current consumption at $U_{\text{cc}} = 5.25\text{ V}$	I	—	—	68	mA
Temperature coefficient of luminous intensity at $\delta a = 25 \dots 85^\circ\text{C}$	-TKI	—	—	1.0	%/K
Reduction coefficient of total at $\delta a = 25 \dots 85^\circ\text{C}$	$-TK_{\text{Pot}}$	—	—	15	mW/K

Brief description

- Four-digit display consisting of four 5×7 -dot LED matrices arranged side by side on a through-hole printed circuit board
- Red-emitting GaAsP chips are used as light emitters.
- Character height 7.5 mm
- The four-digit dot matrices can be arranged in series.
- Preferred applications include data acquisition devices, booking and billing machines, typewriters, NC controls, and small computers

1 Luminous intensity measurement is performed on any diode chip with a beam angle of $15^\circ \pm 3^\circ$

2 $t_p = 250\text{ }\mu\text{s}$, $\tau = 1:10$

3 Inspection by visual check based on reference samples

4 Half-width at half-maximum max. 40 nm

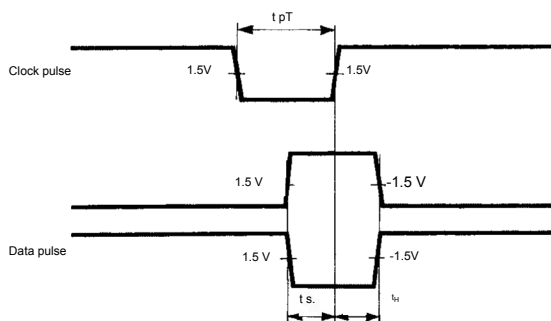


Figure 1: Pulse diagram for driving a VQC 10

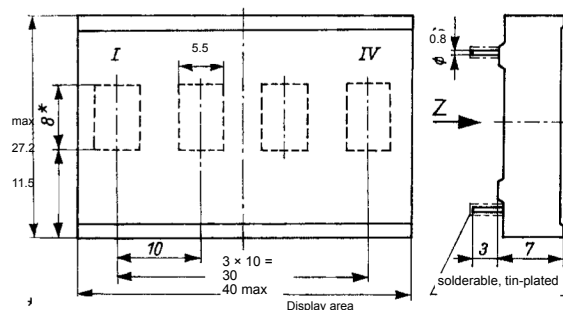


Figure 2: Dimensions of the LED display unit

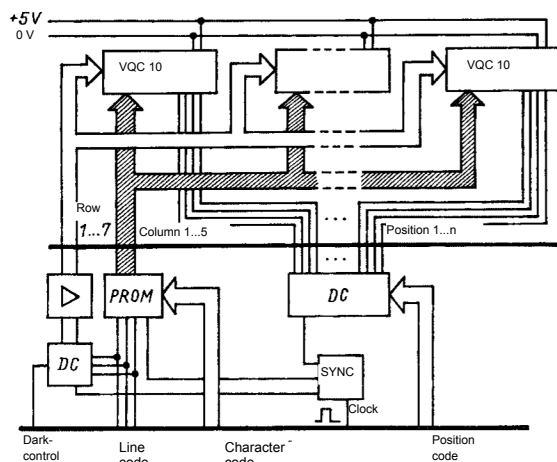
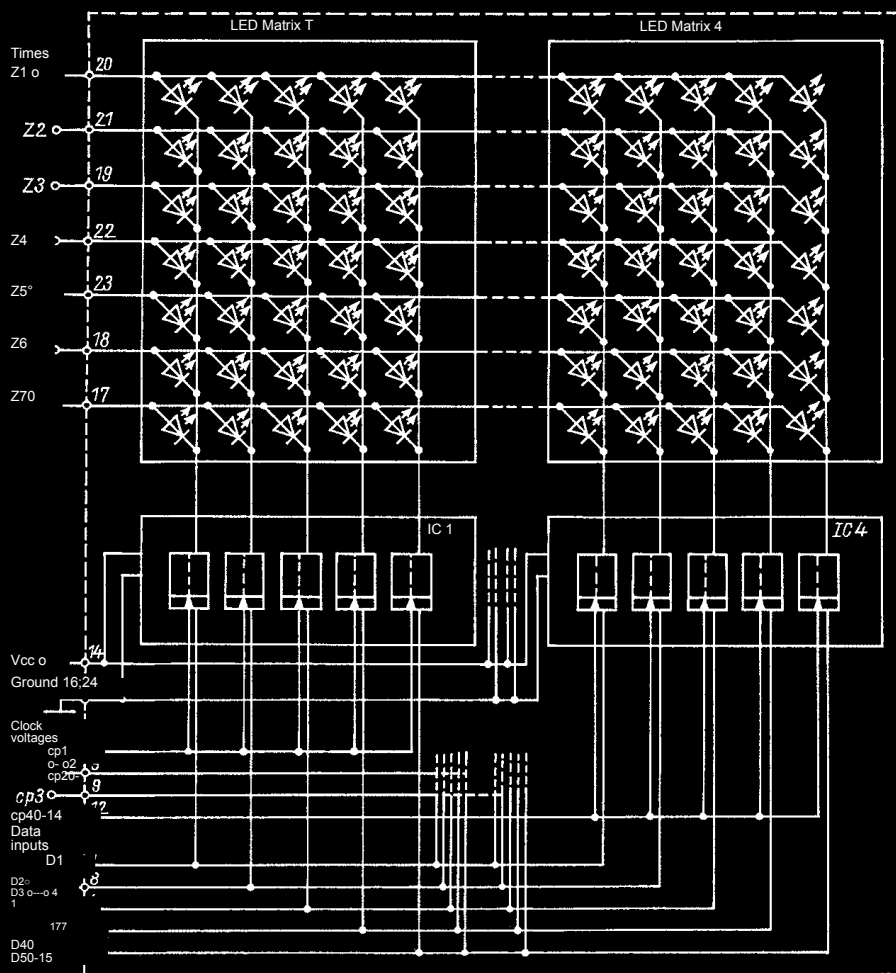


Figure 3: Dimensions and Pin Assignment
(1 D1, 2 cp1, 4 D3, 5 cp2, 7 i. V., 8 D2, 9 cp3, 11 D4, 12 cp4, 14 UCC, 15 D5, 16 Ground, 17 Z7, 18 Z6, 19 Z3, 20 Z1, 21 Z2, 22 Z4, 23 Z5, 24 Ground, 3, 6, 10, 13 Heat sink pins to ground)

Figure 4: Overview schematic for driving a multi-digit alphanumeric 5×7 display



Limit values
($\theta_a = -25 \dots 85^\circ \text{C}$)

- Operating voltage max.
7 V, min. 0 V
- Max. input voltage
5.5 V, min. -0.8 V
- Line input voltage
max. 5 V, min. 0 V
- Total power dissipation (θ_a)
max. 25 $^\circ\text{C}$
at $U_{cc} = 5 \text{ V}$, $U_{IZ} = 5 \text{ V}$,
 $r = 1:8$ and 20 per digit
active pixels
: 1.65 W
- Clock frequency at
 $U_{cc} = 5.25 \text{ V}$: 1.25 MHz

Figure 5: Schematic Diagram
VQC 10

Application

The application circuit for a 16-digit display unit is shown in the adjacent figure. The matrix displays H1 through H4 are interconnected such that a total of seven row, five data, and 16 clock lines

component count, number of connections, and the simplest possible control. The decoder IC D1 is used to drive the row drivers. This makes it possible to activate each row individually or keep all rows inactive via the binary row signals ZA0 through ZA2. The subsequent driver transistors switch the row connections to U_{cc} . Since all corresponding column lines of displays H1 through H4 are combined, IS D2 is used for load matching between the driver and display units. The generation of the 16 clock signals is enabled by a 1-to-16 decoder D3 (74 154). To ensure defined clock pulses, an enable input was used. A frequency of 1 kHz was selected for the row multiplexing. The resulting frame rate of approximately 140 Hz proves to be sufficient.

